

An Open Multi-Station Benchmark for Streamflow Forecasts and Hydrological Drought Indication in the Peruvian Altiplano

Richar Andre Vilca-Solorzano and Dina Maribel Yana-Yucra

Abstract Hydrological variability and drought threaten water security in the Peruvian Altiplano, where sparse gauges and closed tooling limit reproducible forecasting. We release an open multi-station machine-learning benchmark for short-horizon streamflow forecasts and a simplified standardized streamflow drought indicator (SSI-30). Quality-controlled daily discharge from Andean-GC is paired with Open-Meteo/ERA5-Land precipitation across 13 long Peru gauges (≥ 5000 QC days), with Altiplano focus on P00000055 (Puente Isla Cabanillas). Persistence and seasonal climatology are compared with Ridge, random forests, gradient boosting, XGBoost and a small multilayer perceptron under chronological splits and walk-forward checks, reporting RMSE, MAE, NSE, Kling–Gupta efficiency (KGE) and skill versus persistence. Key findings: at a one-day lead, persistence remains strongest (NSE = 0.902); at seven days, Ridge improves on persistence (NSE = 0.550 vs. 0.462; skill = 0.086), and all 13 gauges show positive best-model skill (median = 0.102). This is a transparent Andean research benchmark—not an operational SENAMHI replacement.

Key words: streamflow forecasting, hydrological drought, Altiplano, open data, machine learning, persistence baseline, Nash–Sutcliffe efficiency, Kling–Gupta efficiency, multi-station benchmark

Richar Andre Vilca-Solorzano (corresponding author)

Professional School of Statistical and Computer Engineering, Faculty of Statistical and Computer Engineering (FINESI), National University of the Altiplano (UNAP), Puno, Peru.

e-mail: \emailAndre{}. ORCID: \orcidAndre{}

Dina Maribel Yana-Yucra

Professional School of Statistical and Computer Engineering, Faculty of Statistical and Computer Engineering (FINESI), National University of the Altiplano (UNAP), Puno, Peru.

e-mail: \emailMaribel{}. ORCID: \orcidMaribel{}

1 Introduction

Communities and irrigated agriculture in the Peruvian Altiplano depend on highly seasonal streamflow into the Lake Titicaca system. Climate variability, glacier retreat in headwaters and limited real-time tooling make even short-range discharge forecasts valuable for water allocation and drought awareness. Large-sample machine-learning (ML) hydrology has shown that data-driven models can compete with process models when trained carefully [9, 16], and open catchment datasets such as CAMELS have set a reproducibility standard for multi-basin benchmarking [15]. Yet many published Andean pipelines remain hard to reuse because gauges, licences and code are fragmented.

This paper delivers a laptop-scale, open-data multi-station Peru benchmark with Altiplano focus. Concretely, we:

1. assemble a documented provenance stack—Andean-GC discharge [1] plus Open-Meteo/ERA5-Land precipitation [7, 8]—for 13 Peru gauges with ≥ 5000 quality-controlled (QC) days;
2. enforce leakage-free chronological validation and an expanding-window walk-forward check on the Altiplano focus gauge;
3. report NSE [2], KGE [3, 17] and skill versus persistence [18], so short-lead autocorrelation is not mistaken for model skill;
4. publish a parametric SSI-30 drought indicator (month-wise z -score of the 30-day mean) as a transparent but simplified companion to distributional SSI/SRI practice [10, 19], with open code and figures at <https://github.com/Andre031222/streamflow-drought->

What we do not claim. We do not replace SENAMHI operational forecasts; we do not assert superiority of deep learning over linear models at these leads; and SSI-30 here is not a full Vicente-Serrano distributional SSI product.

This hydrology paper continues an open Altiplano research line by the same team (frost/ $T_{\min}$ multi-station climate ML [6]; AgroYachay cloud IoT/LLM for Andean smallholders [4]; AgroCommish desktop ESP32 commissioning on Zenodo [5], SoftwareX in preparation—not cited as a published SoftwareX article). The present work is the open streamflow/drought-analysis branch of that lineage, not a farm-IoT systems paper.

2 Related Work

Data-driven streamflow models are now common in mountain basins. Random forests have been used for short-lead runoff and flash-flood forecasting in tropical Andean catchments [11, 21]; support-vector regression has been tested for monthly runoff in the Central Andes of Argentina [22]. In Peru, neural networks have been applied to monthly forecasts in high-Andean basins such as Crisnejas [12], and k -nearest neighbours to daily Ramis (Altiplano) flows [13]. Large-sample

Table 1 Compact comparison of related Andean/Peru ML streamflow studies versus this open multi-station benchmark. Entries summarise scope as reported in the cited sources; they are not a meta-analysis of reimplemented scores.

Study	Domain	Lead / scale	Persistence KGE	/ Open multi-station
Muñoz et al. [11, 21]	Tropical Andes (Ecuador); RF	Short lead / one catchment	NSE focus; not Peru	No
Ramos et al. [12]	Crisnejas, Peru; ANN	Monthly	Basin-specific ANN	No
Quispe et al. [13]	Ramis (Altiplano); k -NN	Daily, river	one Single-basin forecast	No
Teverovsky et al. [22]	Korsic Central (Argentina); SVR	Andes Monthly	Mountain runoff ML	No
Kratzert / et al. [9, 16]	Gauch Large-sample LSTM hydrology	CAMELS-style	Strong base-lines; Andean gauges	base- Large-sample (non-Peru) open
This study	Peru GC, focus	Andean- Altiplano $h = 13$ gauges	Daily $h = 1, 7$; NSE, KGE, skill	Yes (open)

LSTM hydrology and multi-timescale rainfall–runoff networks provide methodological context for honest ML baselines [9, 16]. Our prior Altiplano frost/ $T_{\min}$ multi-station climate-ML study [6] shares the comparative-benchmark ethos. Complementary work by Torres-Cruz and coauthors addresses climate vulnerability of agricultural yields in southern Peru with Bayesian spatial (CAR) models [23] and blocked/hierarchical cross-validation under spatial autocorrelation for Peruvian agricultural surveys [24]—methodological cautions we follow by preferring chronological and walk-forward splits over shuffled folds. AgroYachay and AgroCommish [4, 5] are related Andean open-software lineage (farm IoT), not hydrology papers.

Hydrological drought monitoring often relies on standardized indices. The Standardized Precipitation Index (SPI) tradition and its runoff/streamflow analogues (SRI/SSI) motivate transparent, scale-aware drought indication [10, 19]; the SPEI further incorporates atmospheric demand [20]. Our SSI-30 is a simplified month-wise z -score of rolling discharge for the open benchmark—not a full distributional SSI. Reanalysis precipitation comes via Open-Meteo from ERA5-Land [7, 8]. Skill reporting follows NSE, KGE, watershed-model guidelines and classical mean-square skill scores [2, 3, 14, 17, 18].

Table 1 situates this work against prior Andean/Peruvian ML streamflow studies. The distinctive angle is an open multi-station Peru benchmark with explicit persistence and KGE honesty at multiple leads, rather than single-basin peak NSE claims alone.

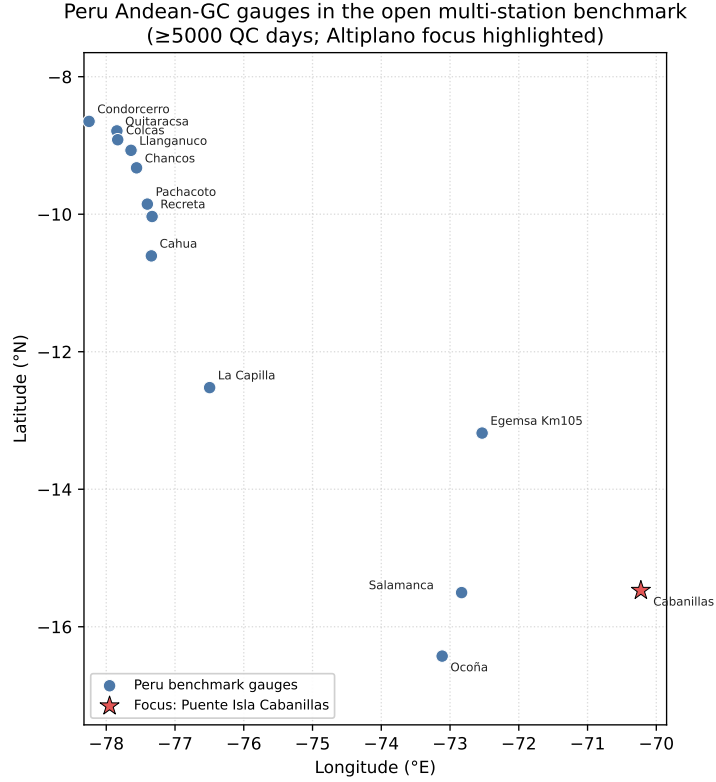


Fig. 1 Locations of the 13 Peru Andean-GC gauges with ≥ 5000 QC days used in the open benchmark. The Altiplano focus station Puente Isla Cabanillas (P00000055) is highlighted.

3 Data and Study Area

We extract daily quality-controlled discharge for all Peru SENAMHI gauges in Andean-GC [1] with $\text{days_w_data_qc} \geq 5000$ ($n = 13$), released on Zenodo under CC BY 4.0. The Altiplano focus station is P00000055 (Puente Isla Cabanillas; approximately 15.47° S, 70.22° W; Coata–Cabanillas / Titicaca system). Southern Peru gauges in the set also include Salamanca and Ocoña. Daily precipitation at each gauge latitude/longitude is obtained from the Open-Meteo historical API (ERA5-Land) [7, 8]. For Cabanillas, the modelling window after feature alignment spans 1994-08-15 to 2024-12-30. Figure 1 maps the 13 gauges; Fig. 2 shows the observed hydrograph with chronological train, validation and test segments.

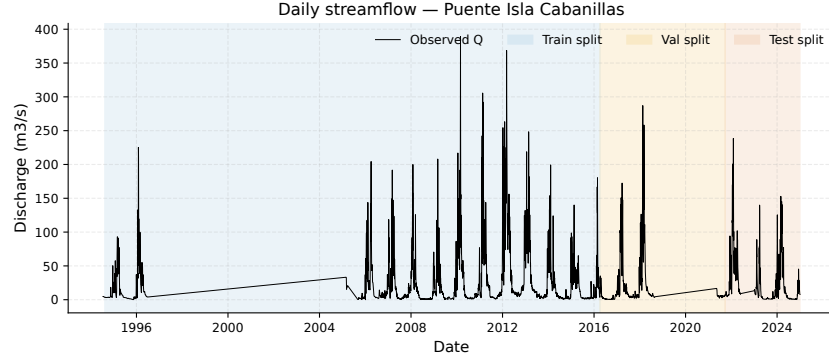


Fig. 2 Daily discharge at Puente Isla Cabanillas with chronological train, validation and test periods.

4 Methods

4.1 Features

For lead times $h \in \{1, 7\}$ days we predict Q_{t+h} from 21 predictors: Fourier seasonality (sin/cos of day-of-year), discharge lags $\{1, 2, 3, 7, 14, 30\}$, rolling means and standard deviations over $\{7, 14, 30\}$ days, and precipitation lags/rolls $\{0, 1, 2, 3, 7\}$ and $\{7, 30\}$.

4.2 Models and baselines

Baselines: persistence ($Q_{t+h} = Q_t$) and seasonal climatology (mean discharge by day-of-year on the training set). Learners: Ridge regression, Random Forest, Gradient Boosting, XGBoost and a small multilayer perceptron. All fits run on a commodity laptop CPU (full multi-station run ≈ 126 s).

4.3 Evaluation

We use a chronological 70/15/15 split (train/validation/test) with no shuffling on every station. Test metrics are RMSE, MAE, NSE, KGE [3] and skill versus persistence,

$$\text{skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}. \quad (1)$$

Positive skill means improvement over persistence. On the focus gauge we additionally run an expanding-window walk-forward protocol (three folds) for Ridge versus persistence. We also compute SSI-30 as the month-wise z -score of the 30-day rolling mean discharge (parametric approximation; Fig. 7). This is not the full Vicente-Serrano distributional transform and should not be treated as an operational SSI product [10].

5 Results

Four results structure the narrative (all numbers from the chronological test / walk-forward runs summarised in `cifras.tex`):

1. **$h = 1$: persistence wins.** At Cabanillas, persistence NSE = 0.902 (KGE = 0.951) exceeds every learner; best ML is MLP (NSE = 0.871) but remains skill-negative on RMSE.
2. **$h = 7$: Ridge gains modest skill.** Ridge NSE = 0.550 vs. persistence 0.462 (skill = 0.086; KGE = 0.674); MLP is close behind.
3. **Multi-station:** best ML skill is positive at all 13/13 gauges (median skill = 0.102); Ridge alone is skill-positive everywhere (mean skill = 0.078).
4. **Walk-forward:** expanding-window Ridge skill at $h = 7$ is positive in all three Cabanillas folds (mean skill = 0.100).

5.1 Focus station (Cabanillas)

Table 2 details the held-out chronological test period at Puente Isla Cabanillas (2021-09-24–2024-12-30; $n_{\text{test}} = 982$). Figures 3–4 show the $h = 7$ forecast track and skill bars.

5.2 Multi-station Peru benchmark

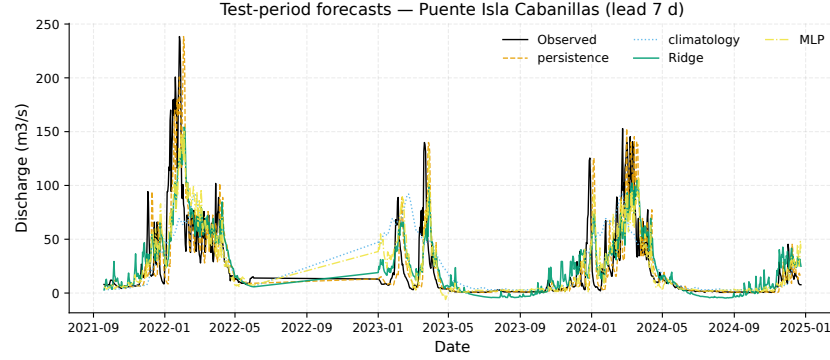
At $h = 7$, Ridge mean NSE = 0.639 versus persistence mean NSE = 0.569 (mean KGE = 0.714). Figures 5–6 summarise cross-station skill and NSE; Fig. 7 shows the parametric SSI-30 series at Cabanillas.

6 Discussion

The one-day result is intentional honesty: strong autocorrelation makes “ML superiority” without a persistence baseline misleading [17, 18]. The seven-day Ridge/MLP

Table 2 Test-set metrics at Puente Isla Cabanillas (chronological split). Skill is relative to persistence.

Model	h	RMSE	MAE	NSE	KGE	Skill
Persistence	1	11.07	4.69	0.902	0.951	0.000
Climatology	1	28.47	15.98	0.353	0.506	−1.571
Ridge	1	13.84	8.08	0.847	0.895	−0.250
RandomForest	1	13.32	6.51	0.858	0.905	−0.203
GradientBoosting	1	12.82	6.63	0.869	0.896	−0.158
MLP	1	12.69	6.55	0.871	0.912	−0.146
XGBoost	1	12.95	6.50	0.866	0.901	−0.169
Persistence	7	25.96	12.57	0.462	0.731	0.000
Climatology	7	28.50	15.93	0.352	0.509	−0.098
Ridge	7	23.74	14.28	0.550	0.674	0.086
RandomForest	7	27.63	14.60	0.391	0.626	−0.064
GradientBoosting	7	25.64	13.95	0.475	0.639	0.012
MLP	7	24.45	13.40	0.523	0.679	0.058
XGBoost	7	27.16	14.74	0.412	0.626	−0.046

**Fig. 3** Observed versus forecast discharge on the Cabanillas test set at a 7-day lead.

gain is modest yet reproducible across all 13 gauges, obtained with open data and about 126 s of laptop CPU. Walk-forward folds support the chronological $h = 7$ skill signal.

Limitations: (i) a single reanalysis precipitation source (Open-Meteo/ERA5-Land), not local SENAMHI rain gauges; (ii) no routing or process constraints; (iii) parametric SSI-30 rather than a full distributional SSI [10, 19]; (iv) heterogeneous record eras (Cordillera Blanca historical vs. modern Altiplano); (v) no formal uncertainty quantification or operational assimilation. Natural extensions include more Altiplano gauges when QC length allows, distributional SSI/SPEI products [20], and LSTM/transfer learning following large-sample practice [9, 16]. This work does not replace SENAMHI or institutional warning systems.

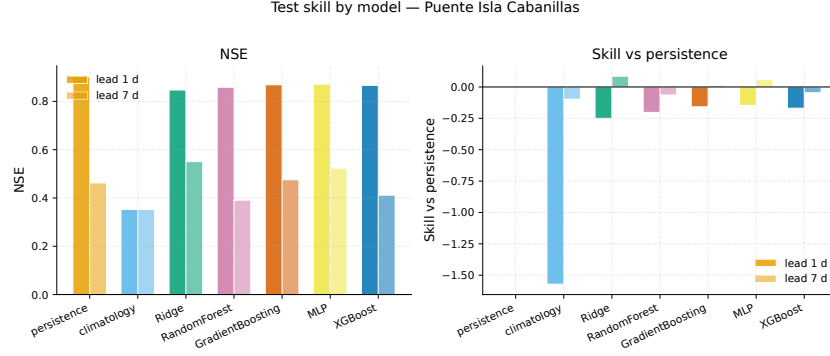


Fig. 4 Skill versus persistence by model and lead time at Puente Isla Cabanillas. Positive values favour the learner.

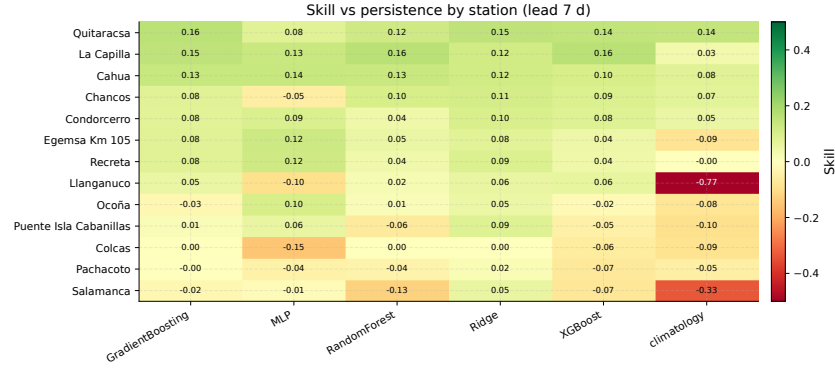


Fig. 5 Multi-station skill versus persistence at a 7-day lead (Peru Andean-GC gauges with ≥ 5000 QC days).

7 Conclusion

Main takeaway. An open, laptop-scale Peru multi-station benchmark with Altiplano focus shows that persistence remains the strongest one-day baseline (Cabanillas $NSE = 0.902$), while lightweight Ridge models deliver modest but consistent seven-day skill over persistence on all 13 long gauges (focus skill $= 0.086$; multi-station median best-ML skill $= 0.102$), with walk-forward confirmation (mean skill $= 0.100$).

Limitations. Results rest on a single precip source, a parametric SSI-30 (not operational distributional SSI), and research-grade chronological splits—not a SENAMHI replacement.

Future work. Expand the Altiplano gauge set when QC length allows; adopt distributional SSI/SPEI; and test LSTM/transfer learning under the same persistence-honest protocol. Full code and metrics: <https://github.com/Andre031222/streamflow-drought-alti>

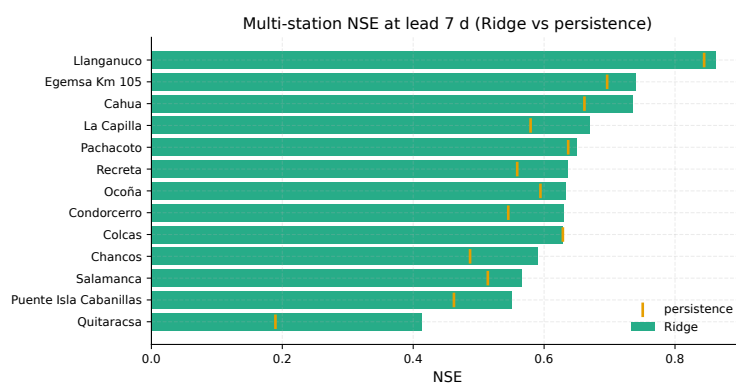


Fig. 6 Multi-station NSE at a 7-day lead for Ridge versus persistence markers.

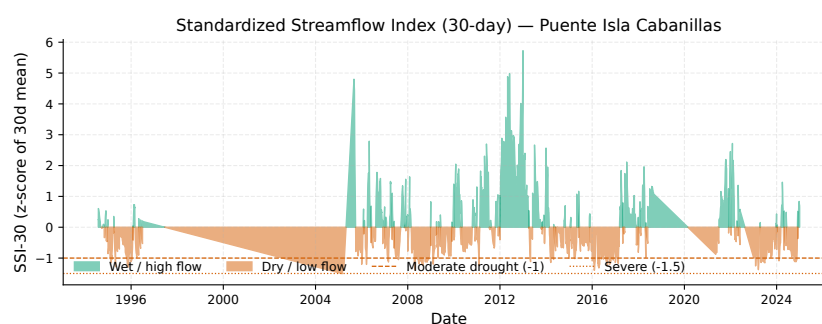


Fig. 7 SSI-30 at Puente Isla Cabanillas: month-wise z-score of the 30-day rolling mean (parametric approximation).

Data availability

Daily QC discharge is from Andean-GC v1.1 on Zenodo (CC BY 4.0; doi:<https://doi.org/10.5281/zenodo.18035801>). Daily precipitation was retrieved from the Open-Meteo Historical Weather API (ERA5-Land). Station extracts, precip caches and processed tables are documented in the project inventory file `INVENTARIO.md` under `02-datos/fuentes/`.

Code availability

Analysis code (feature construction, models, metrics, figures) is provided under an MIT license in the public GitHub repository: <https://github.com/Andre031222/streamflow-drought-benchmark>. Reproduce with `python 03-codigo/run_experiment.py` after installing `requirements.txt`.

Author contributions

R.A.V.-S.: conceptualization, data curation, methodology, software, formal analysis, visualization, writing—original draft. D.M.Y.-Y.: methodology, validation, writing—review and editing. Both authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethics and limitations

No human subjects or personal data were used. Streamflow and reanalysis fields are public open data. Forecasts are research benchmarks only and must not be used as sole input for flood or drought emergency decisions without institutional (e.g. SENAMHI) validation.

Acknowledgements

The authors thank open-data providers Andean-GC and Open-Meteo/Copernicus, and Universidad Nacional del Altiplano (UNAP) / FINESI for academic affiliation. AgroYachay and AgroCommish are cited only as related Andean open-source software lineage; this conference paper is an independent open hydrology benchmark.

References

1. R. Aguayo, H. Zekollari, M. van Tiel, J. Bolibar, L. Van Tricht, A. I. Ayala Ramos, L. Ultee. Andean Glacierized Catchment (Andean-GC) dataset (Version v1.0). Zenodo (2026). doi:<https://doi.org/10.5281/zenodo.18035801>
2. J. E. Nash, J. V. Sutcliffe. River flow forecasting through conceptual models part I—A discussion of principles. *Journal of Hydrology* **10**(3), 282–290 (1970). doi:[https://doi.org/10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6)
3. H. V. Gupta, H. Kling, K. K. Yilmaz, G. F. Martinez. Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of Hydrology* **377**(1–2), 80–91 (2009). doi:<https://doi.org/10.1016/j.jhydrol.2009.08.003>
4. F. Torres-Cruz, R. A. Vilca-Solorzano, D. M. Yana-Yucra, V. Ibañez-Quispe, E. L. Fuentes-Navarro. AgroYachay: An open-source IoT and large-language-model platform supporting agronomic and economic decision-making for Andean smallholders. *SoftwareX* **36**,

- 103037 (2026). doi:<https://doi.org/10.1016/j.softx.2026.103037>. Source code: <https://github.com/Andre031222/agroyachay>; Zenodo: doi:<https://doi.org/10.5281/zenodo.20829992>
5. R. A. Vilca-Solorzano, D. M. Yana-Yucra, R. Quispe-Vargas, V. Ibañez-Quispe, F. Torres-Cruz. AgroCommish: A Plug-and-Play Desktop Tool for End-to-End Manufacturing and Commissioning of ESP32-Based Agricultural IoT Sensor Nodes. Zenodo software release (2026). doi:<https://doi.org/10.5281/zenodo.20655609>. Source code: <https://github.com/Andre031222/agrocommish>. Note: open software archive; *SoftwareX* manuscript in preparation (not a published journal article).
 6. F. Torres-Cruz, D. M. Yana-Yucra, R. A. Vilca-Solorzano. Comparative Analysis of Statistical, Machine Learning, and Deep Learning Approaches for Frost Prediction in the High Andean Region of Peru. *International Journal of Advanced Computer Science and Applications (IJACSA)* **16**(9) (2025). doi:<https://doi.org/10.14569/IJACSA.2025.0160992>. Open code/data: <https://github.com/Andre031222/frost-prediction-altiplano-puno>
 7. H. Hersbach et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* **146**(730), 1999–2049 (2020). doi:<https://doi.org/10.1002/qj.3803>
 8. J. Muñoz-Sabater et al. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* **13**, 4349–4383 (2021). doi:<https://doi.org/10.5194/essd-13-4349-2021>
 9. F. Kratzert, D. Klotz, G. Shalev, G. Klambauer, S. Hochreiter, G. Nearing. Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences* **23**, 5089–5110 (2019). doi:<https://doi.org/10.5194/hess-23-5089-2019>
 10. S. M. Vicente-Serrano, J. I. López-Moreno, S. Beguería, J. Lorenzo-Lacruz, C. Azorin-Molina, E. Morán-Tejeda. Accurate computation of a streamflow drought index. *Journal of Hydrologic Engineering* **17**(2), 318–332 (2012). doi:[https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000433](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000433)
 11. P. Muñoz, D. F. Muñoz, J. Orellana-Alvear, R. Céleri. Enhancing runoff forecasting through the integration of satellite precipitation data and hydrological knowledge into machine learning models. *Natural Hazards* **121**, 3915–3937 (2025). doi:<https://doi.org/10.1007/s11069-024-06939-w>
 12. C. Ramos et al. Predicción de caudales mensuales en ríos de cuencas altoandinas con enfoque de redes neuronales artificiales. Caso: río Crisnejas, Perú. *Tecnología y Ciencias del Agua* **14**(1) (2023). doi:<https://doi.org/10.24850/j-tyca-14-01-04>
 13. W. Quispe et al. Hydrological modeling based on the KNN algorithm: An application for the forecast of daily flows of the Ramis river, Peru. *Tecnología y Ciencias del Agua* **14**(2) (2023). doi:<https://doi.org/10.24850/j-tyca-14-02-05>
 14. D. N. Moriasi, J. G. Arnold, M. W. Van Liew, R. L. Bingner, R. D. Harmel, T. L. Veith. Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. *Transactions of the ASABE* **50**(3), 885–900 (2007). doi:<https://doi.org/10.13031/2013.23153>
 15. N. Addor, A. J. Newman, N. Mizukami, M. P. Clark. The CAMELS data set: catchment attributes and meteorology for large-sample studies. *Hydrology and Earth System Sciences* **21**, 5293–5313 (2017). doi:<https://doi.org/10.5194/hess-21-5293-2017>
 16. M. Gauch, F. Kratzert, D. Klotz, G. Nearing, J. Lin, S. Hochreiter. Rainfall–runoff prediction at multiple timescales with a single Long Short-Term Memory network. *Hydrology and Earth System Sciences* **25**, 2045–2062 (2021). doi:<https://doi.org/10.5194/hess-25-2045-2021>
 17. W. J. M. Knoben, J. E. Freer, R. A. Woods. Technical note: Inherent benchmark or not? Comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. *Hydrology and Earth System Sciences* **23**, 4323–4331 (2019). doi:<https://doi.org/10.5194/hess-23-4323-2019>

18. A. H. Murphy. Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review* **116**(12), 2417–2424 (1988). doi:[https://doi.org/10.1175/1520-0493\(1988\)116<2417:SSBOTM>2.0.CO;2](https://doi.org/10.1175/1520-0493(1988)116<2417:SSBOTM>2.0.CO;2)
19. S. Shukla, A. W. Wood. Use of a standardized runoff index for characterizing hydrologic drought. *Geophysical Research Letters* **35**, L02405 (2008). doi:<https://doi.org/10.1029/2007GL032487>
20. S. M. Vicente-Serrano, S. Beguería, J. I. López-Moreno. A multiscalar drought index sensitive to global warming: the Standardized Precipitation Evapotranspiration Index (SPEI). *Journal of Climate* **23**(7), 1696–1718 (2010). doi:<https://doi.org/10.1175/2009JCLI2909.1>
21. P. Muñoz, J. Orellana-Alvear, P. Willems, R. Céleri. Flash-flood forecasting in an Andean mountain catchment—development of a step-wise methodology based on the random forest algorithm. *Water* **10**(11), 1519 (2018). doi:<https://doi.org/10.3390/w10111519>
22. S. A. Teverovsky Korsic, C. Notarnicola, M. Uriburu Quirno, L. J. Cara Ramirez. Assessing a data-driven approach for monthly runoff prediction in a mountain basin of the Central Andes of Argentina. *Environmental Challenges* **10**, 100680 (2023). doi:<https://doi.org/10.1016/j.envc.2023.100680>
23. A. J. Maquera Andrade, F. Torres Cruz. Bayesian Spatial Analysis of Agricultural Yields and Climate Vulnerability in Southern Peru Using Hierarchical CAR Models, 2024. In: *Green Technologies and Sustainable Intelligence—Volume II*. Springer Proceedings in Physics (2026). doi:https://doi.org/10.1007/978-3-032-20151-5_4
24. W. Vilca Apaza, F. Torres Cruz, E. E. Carpio Vargas. Blocked and Hierarchical Cross-Validation for Agricultural Data with Spatial Autocorrelation: Application to Peru's National Agricultural Survey. In: *Green Technologies and Sustainable Intelligence—Volume II*. Springer Proceedings in Physics (2026). doi:https://doi.org/10.1007/978-3-032-20151-5_1